Last Name: First Name:

Matriculation Number:

Exam
Artificial Intelligence 1

April 9, 2025

Please ignore the QR codes; do not write on them, they are for grading support

	To be used for grading, do not write here												
prob.	1.1	2.1	2.2	3.1	4.1	5.1	5.2	5.3	6.1	7.1	7.2	Sum	grade
total	10	8	10	7	3	6	8	6	8	12	7	85	
reached													

Organizational Information

Please read the following directions carefully and acknowledge them with your signature.

1. Please place your student ID card and a photo ID on the table for checking.
2. You can reach 85 points if you fully solve all problems. You will only need 85 points for a perfect score, i.e. 0 points are bonus points.
3. No resources or tools are allowed except for a pen.
4. You have 90 min (sharp) for the exam.
5. Write the solutions directly on the sheets, no other paper will be graded.
6. If you have to abort the exam for health reasons, your inability to sit the exam must be certified by an examination at the University Hospital. Please notify the exam proctors and have them give you the respective form.
7. Please make sure that your copy of the exam is complete (5 pages excluding cover sheet and organizational information pages) and has a clear print. **Do not forget to add your personal information on the cover sheet and to sign this declaration.**

Declaration: With my signature I certify having received the full exam document and having read the organizational information above.

Erlangen, April 9, 2025 .....
(signature)

Organisatorisches

Bitte lesen die folgenden Anweisungen genau und bestätigen Sie diese mit Ihrer Unterschrift.

1. Bitte legen Sie Ihren Studierendenausweis und einen Lichtbildausweis zur Personenkontrolle bereit!
2. Sie können 85 Punkte erreichen, wenn Sie alle Aufgaben vollständig lösen. Allerdings zählen 85 Punkte bereits als volle Punktzahl, d.h. 0 Punkte sind Bonuspunkte.
3. Es sind keine Hilfsmittel erlaubt außer einem Stift.
4. Die Bearbeitungszeit beträgt genau 90 min.
5. Schreiben Sie die Lösungen direkt auf die ausgeteilten Aufgabenblätter. Andere Blätter werden nicht bewertet.
6. Wenn Sie die Prüfung aus gesundheitlichen Gründen abbrechen müssen, so muss Ihre Prüfungsunfähigkeit durch eine Untersuchung in der Universitätsklinik nachgewiesen werden. Melden Sie sich in jedem Fall bei der Aufsicht und lassen Sie sich das entsprechende Formular aushändigen.
7. Überprüfen Sie Ihr Exemplar der Klausur auf Vollständigkeit (5 Seiten exklusive Deckblatt und Hinweise) und einwandfreies Druckbild! **Vergessen Sie nicht,**

auf dem Deckblatt die Angaben zur Person einzutragen und diese Erklärung zu unterschreiben!

Erklärung: Durch meine Unterschrift bestätige ich den Empfang der vollständigen Klausurunterlagen und die Kenntnisnahme der obigen Informationen.

Erlangen, April 9, 2025 .....
(Unterschrift)

Please consider the following guidelines to avoid losing points:

- If you continue an answer on another page, clearly give the problem number on the new page and a page reference on the old page.
- You can always ask for the translation or explanation of a non-technical word.
- If you do not want something to be graded, clearly cross it out. Adding a wrong statement to a correct solution may lead to deductions.
- The instructions "Give X", "List X" or similar mean that only X is needed. If you additionally justify your answer, we may or may not give you partial credit for a wrong answer.
- The instruction "Assume X" means that X is information that you may use in your answer.
- The instruction "Model X as a Y" means that you have to describe X formally and exactly as an instance of Y using the definition of Y from the lecture.
- If you are uncertain how long or complex an answer should be, use the number of points as an indication: 1 point roughly corresponds to 1 minute.
- In all calculation questions, you have to simplify as much as reasonably possible without a calculator. For example, $\log 2$ or 3^7 should not be calculated, but $0.4 \cdot 0.3 \cdot 0.5 = 0.06$ should be.

1 Prolog

Problem 1.1 (Interpreting and Writing Clauses)
Consider the following sequence of Prolog clauses:

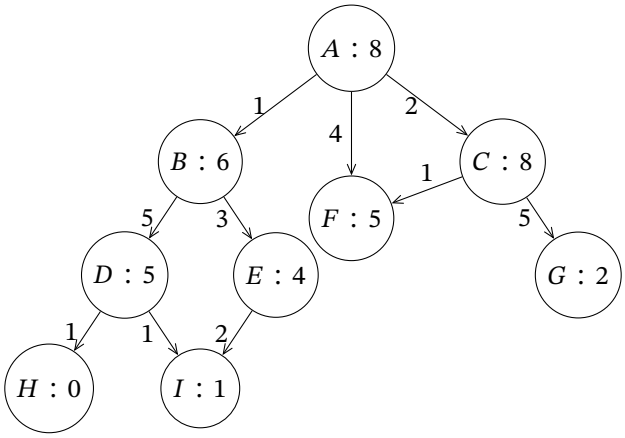
```
descendant(D,X) :- descendant(D,P), parent(P,X).
descendant(D,X) :- parent(D,X).
sibling(X,Y) :- parent(P,X), parent(P,Y).
sibling(william,harry).
parent(elizabeth,charles).
parent(charles,william).
```

Below, you may abbreviate all names by their first letter.

1. Give all function and predicate constants and their arities that make up the first-order signature underlying these clauses. 4 Points
2. Give a clause that can be used to extend the above with the definition of a binary predicate for the niece-or-nephew relation. 2 Points
3. Interpreting the clauses as a Prolog program, what is output when we execute the query `descendant(X,Y)`? 2 Points
4. Interpreting each clause as a hypothesis in first-order predicate logic, which instances of the formula `descendant(X,Y)` can be derived? 2 Points

2 Search

Problem 2.1 (Search Algorithms)
Consider the graph below. Every edge is labeled with its cost.
Each node is labeled with $n : h(n)$ where n is its name and $h(n)$ its value for heuristic h.



Assume you have already expanded the root node A. In each problem below, list the **next** 4 nodes that will be expanded using the respective search algorithm and (if needed) heuristic h.
If there is a tie, break it using alphabetical order.

1. depth-first search 1 Points
2. breadth-first search 1 Points
3. uniform-cost search 2 Points
4. greedy search 2 Points

5. A^*-search 2 Points

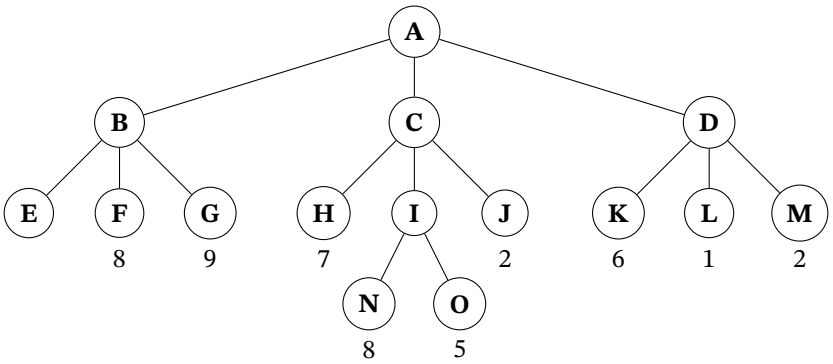
Problem 2.2 (Search Problems)
Fix the set $V = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.
Consider the family of deterministic search problems $P_n = \langle S, A, T, I, G_n \rangle$ for $n = 0, 1, 2, ...$ given by

- S is the set of finite sequences over V.
- $A = \{b\} \cup \{a_i \mid i \in V\}$
- T is given by
  - $T(a_i, s) = \{s'\}$ where s' is s with i appended at the end
  - $T(b, s) = \begin{cases} \emptyset & \text{if } s \text{ is the empty sequence} \\ \{s'\} & \text{otherwise, where } s' \text{ is } s \text{ without its last element} \end{cases}$
- I is the set containing the empty sequence.
- G_n is the set of all sequences of distinct elements that sum to n.
  For example, G_3 contains (among others) 3 and 0, 3 and 2, 0, 1 but not 1, 1, 1 (whose elements are not distinct) or 1, 3 (whose elements do not sum to 3).

1. Give the result of applying the action sequence a_0, a_1, b, a_2 in the initial state. 2 Points
2. Assume BFS tries the actions in the order $a_0, ..., a_9, b$. Which solution will it find first for P_{22}? 3 Points
3. Give a value of n for which P_n has no solution. 2 Points
4. First explain the key reason why informed search strategies will typically perform better on this particular problem than uninformed ones; then give a useful heuristic for P_n. 3 Points

3 Adversarial Search

Problem 3.1 (Minimax)
Consider the following minimax game tree for the **maximizing** player's turn. The values at the leaves are the static evaluation function values of those states; some of those values are currently missing.



1. Label the nodes I and C with their minimax value. 2 Points
2. Give every possible label for the node **E** that would result in the player definitely choosing move **B** (no matter how ties are broken). 2 Points
3. Now assume **E** is labeled with 5, and we use $\alpha\beta$-pruning. We expand child nodes in alphabetical order. Which nodes would be pruned? 3 Points

4 Constraint Satisfaction

Problem 4.1 (Basic Definitions)
Consider the following constraint network $\langle V, D, C\rangle$

- $V = \{a, b, c, d\}$
- $D_a = \texttt{bool}$, $D_b = D_c = \{0, 1, 2, 3\}$, $D_d = \{0, 1, 2, 3, 4, 5, 6\}$
- Constraints C:
  - if a, then $b \leq 2$
  - if $c < 2$, then a
  - $b + c < 4$
  - $b > d$
  - $d = 2c$

1. Give all solutions. 1 Points
2. Give an inconsistent total variable assignment. 1 Points
3. Give all consistent partial assignments α such that $\mathrm{dom}(\alpha) \subseteq \{a, b\}$. 0 Points
4. Assume a CSP algorithm starts by assigning *true* to a. How do the domains of b, c, d change after forward checking? 1 Points

5 Logic

Problem 5.1 (Propositional Logic)

1. Fix the signature with propositional variables p, q, r. Give a formula that has exactly 1 satisfying assignment. 2 Points
2. Explain the Unit Propagation rule that is used in DPLL solvers for SAT. 2 Points
3. Assume C is in CNF, explain how you can get a formula in DNF that is equivalent to $\neg C$. 2 Points

Problem 5.2 (Predicate Logic)
Consider the following signature of predicate logic:

- unary function constant f
- binary function constant g
- binary predicate constant p

1. Give a formula that uses all three constants and is a theorem. 2 Points
2. Now consider the formulae A and B given by 3 Points

$$A = \forall x.\forall y.p(f(x), g(x, y))$$

$$B = \forall x.\forall y.p(x, y)$$

   Give a model $\langle \mathcal{D}, \mathcal{I}\rangle$ that satisfies A but falsifies B.
3. Prove or refute the following statement: 3 Points
   If the formula A has one free variable x, and there is a model in which A is true under all assignments, then A is valid.

Problem 5.3 (Proving)
Consider first-order predicate logic with unary predicate symbols P and Q.

1. Prove the following formula using tableaux calculus: 6 Points

$$(\forall x.P(x) \Rightarrow \neg Q(x)) \Rightarrow ((\exists y.P(y)) \Rightarrow \neg\forall z.Q(z))$$

6 Knowledge Representation

Problem 6.1 (ALC)
Consider the following description logic signature

- concept symbols: m (for male), f (for female)
- role symbols: p (for parent-of), s (for sibling-of)

It is meant to model the world as indicated in the brackets.

1. Give an ABox for this signature that declares a woman E, her son C, and his sons W and H. 2 Points
2. Give a concept describing the set of childless men who have a niece. 2 Points
3. ALC is not good at expressing numeric constraints such as forming the concept of people with at least **two** children. Assume we add a production $F ::= \exists^n R.F$ to the grammar that behaves like $\exists$ but requires at-least-n elements instead of at-least-one. 2 Points
   Give an appropriate definition of the interpretation of $\exists^n R.F$ in a model $\langle \mathcal{D}, [\![-]\!] \rangle$.
4. Give the translation to first-order logic of the axiom $(\forall p.f) \sqsubseteq (\exists s.m)$ 2 Points

7 Planning

Problem 7.1 (Planning Deliveries in STRIPS)
Consider a truck that can carry 4 **objects at a time** and is supposed to deliver objects $Obj = \{V, W, X, Y, Z\}$ from location A to certain locations $Loc = \{A, B, C, D\}$ along some roads

$$Roads = \{(A,B),(B,A),(B,C),(C,B),(B,D),(D,B)\}.$$

We use the following STRIPS task:

- facts: $\{\texttt{at}(l,o) \mid l \in Loc, o \in Obj\} \cup \{\texttt{truck}(l) \mid l \in Loc\}$
- the action $\texttt{move}(l,m,O)$ delivers up to 4 objects from l to m: for $(l,m) \in Roads$ and $O \subseteq Obj$, and $\#(O) \leq 4$, it is given by
  - precondition: $\texttt{at}(l,o)$ for all $o \in O$, $\texttt{truck}(l)$
  - add list: $\texttt{at}(m,o)$ for all $o \in O$, $\texttt{truck}(m)$
  - delete list: same as precondition
- initial state: $\texttt{truck}(C)$, $\texttt{at}(A,o)$ for $o \in Obj$
- goal state: $\texttt{at}(C,V)$, $\texttt{at}(D,W)$, $\texttt{at}(B,X)$, $\texttt{at}(B,Y)$, $\texttt{at}(A,Z)$

1. Give a sequence of two actions that is applicable in the initial state and give the resulting state. 4 Points
2. Give an optimal plan for the task above. 4 Points
3. Consider the following heuristic: $h(s) = \frac{1}{4}\sum_{o \in Obj} d(s,o)$ where $d(s,o)$ is the number of roads separating the location of o in state s from its location in the goal state. Argue whether this heuristic is admissible or not. 4 Points

Problem 7.2 (Search as Planning)
We saw in the course that planning is a special case of search. Complete the argument below establishing that the converse is also true.

1. Given a search problem $\langle S, A, T, I, G \rangle$, we define a planning task $\langle P', A', I', G' \rangle$ as follows: 7 Points
   - P' contains the facts *start* and *inGoal* as well as one fact $in(\square)$ for every $s \in \square$.

- A' contains the following actions, all given as triples $\langle pre, add, del\rangle$ of preconditions, add-list, and delete-list:
  - $init(s)$ of the form $\langle\{start\},\{in(s)\},\{start\}\rangle$ for every $s \in I$
  - $goal(s)$ of the form $\langle\{in(s)\},\{inGoal\},\{in(s)\}\rangle$ for every $s \in G$
  - $do(s,a,s')$ of the form $\langle in(s),$ [ ], [ ]$\rangle$ for every $s \in S$, every $a \in A$, and every $s' \in$ [ ]
- $I' = \{start\}$
- $G' = \{inGoal\}$

Now we have a bijection between

- solutions $a_1,\ldots,a_n$ (visiting the states $s_0,\ldots,s_n$) for the search problem, and
- plans of the form

  [ ]

  that [ ] the planning task.